

Eigenvalues, Eigenvectors, and Characteristic Polynomial

% Produce a column vector E containing the eigenvalues of a square matrix A.

```
A = [1,2; 5,4]; E = eig(A)
```

% Produce a diagonal matrix D of eigenvalues and a full matrix V whose columns are the corresponding eigenvectors so that $A*V = V*D$.

```
A = [1,2;5,4]; [V, D] = eig(A)
```

% Generation of eigenvalues and eigenvectors of a square matrix of order 3

```
A = randi([-5,5], 3)
```

```
[V, D] = eig(A)
```

```
Lambda1 = D (1, 1)
```

```
Lambda2 = D (2, 2)
```

```
Lambda3 = D (3, 3)
```

```
Eigenvector1 = V (: 1)
```

```
Eigenvector2 = V (: 2)
```

```
Eigenvector3 = V (: 3)
```

% Produce a diagonal matrix D of eigenvalues and a full matrix R whose columns are the corresponding right eigenvectors (eigenvectors) and also produces a full matrix L whose columns are the corresponding left eigenvectors so that $L'*A = D*L'$

```
A = randi([-5,5],3); [R,D,L] = eig(A)
```

% Obtain characteristic polynomial of a matrix A.

```
A = [1 2 3; 2 3 4; 3 5 8];
```

```
syms x;
```

```
charpoly(A, x)
```

% Find eigenvalues of a magic square matrix of order 9.

```
M=magic(9); eig(M)
```

NOTE: magic(N) is an N-by-N matrix constructed from the integers 1 through N² with equal row, column, and diagonal sums, produces valid magic squares for all N > 0 except N = 2.

% Generate a matrix A with given eigenvalues Ld1, Ld2, Ld3 and independent eigenvectors Ev1, Ev2 and Ev3.

```
Ld1 = 1;
```

```
Ld2 = 2;
```

```
Ld3 = 3;
```

```
Ev1 = [1;1;1];
```

```
Ev2 = [2;1; -1];
```

```
Ev3 = [1;2; -1];
```

```
V = [Ev1, Ev2, Ev3]
```

```
D = diag([Ld1, Ld2,Ld3])
```

```
A = V*D*inv(V)
```

Practise Questions

1. Generate random integer square matrices A of order 4, B of order 10 and C of order 15. Find their eigenvalues and eigenvectors.
 - a. Verify the following properties of eigenvalues for A , B and C :
 - i. Trace of a matrix = sum of eigenvalues of the matrix
 - ii. Det of a matrix = Product of eigenvalues of the matrix
 - iii. Complex eigenvalues of a matrix appear in pairs
 - b. Find eigenvalues of $5A$, $B/3$ and C^{-1} . Compare them with eigenvalues of A , B and C respectively.
 - c. Find eigenvectors of $5A$, $B/3$ and C^{-1} . Compare them with eigenvectors of A , B and C respectively.
 - d. From A , using $S = A + A'$, generate a symmetric matrix. Find the eigenvalues and eigenvectors of S .
2. Generate a random integer square matrix A of order 3.
 - a. Find the eigenvalues and eigenvectors of this matrix.
 - b. Find the characteristic polynomial for this matrix.
 - c. Replace x in the characteristic polynomial by the matrix A and evaluate the value. What is the value obtained?
3. Generate an integer matrix A of order 5×3 . Find AA^T and $A^T A$. Evaluate the eigenvalues of AA^T and $A^T A$. Verify the result that **“The non-zero eigenvalues of AA^T and $A^T A$ are the same”**.
4. Generate a matrix with eigenvalues as 3, 5, 7, 11 and eigenvectors as $[1, 1, 2, 2]$, $[1, 0, 1, 0]$, $[2, 1, 2, 0]$ and $[0, 1, 1, 0]$.
5. Using MATLAB, generate a 5×5 matrix A of rank 2.
 - a. Obtain a symmetric matrix $B = A + A'$ and find the rank of B .
 - b. Find the eigenvalues and eigenvectors of B using MATLAB.
 - c. Verify if the eigenvectors of B are orthogonal.
6. Using MATLAB, generate a 10×5 matrix A of rank 3.
 - a. Obtain a symmetric matrix $S_1 = A * A^T$ and $S_2 = A^T * A$
 - b. Find the rank of S_1 and S_2 .
 - c. Find the eigenvalues of S_1 and S_2 .
7. Using MATLAB, generate a 7×7 matrix A of rank 2.
 - a. Obtain a skew-symmetric matrix $B = A - A'$ and find the rank of B .
 - b. Find the eigenvalues and eigenvectors of B using MATLAB.
 - c. In a similar manner, generate a skew-symmetric matrix C from an 8×8 matrix. Does C have a zero eigenvalue?
8. Generate a matrix that will transform the points on a circular membrane to an elliptic membrane with a stretch factor of 5 and 9 along the directions $[1, 1]$ and $[1, -1]$ respectively.